

# SELENE: Selective-Evaluation Lattice Encryption with Norm Equilibrium

## Module-LWE Inner-Product Functional Encryption -- A Construction Outside the GPV-CKKS Impossibility Regime

Trapdoor-Free Key Derivation -- RLWE-Small Secrets -- IND-FE from MLWE -- Explicit Novelty over ALS16/BJKL18  
QuantumVault Research Group -- Blind Review Copy

### Abstract

Post-quantum functional encryption requires schemes secure under lattice hardness and free of large-key constraints from trapdoor sampling. A companion impossibility [QV-Thesis, Theorem 8.1] shows GPV-adaptive HIBE and CKKS-FHE in a shared ring produce an irreconcilable key-norm collision; SELENE is motivated by this result but operates outside its scope by design. We present MLWE-IPFE achieving identity-based slot access and linear evaluation over encrypted data -- without SampleLeft, CKKS arithmetic, or shared ring constraints. Key derivation:  $sk_f = \sum_i f_i s_i$  (linear combination of RLWE-small secrets), giving  $\|sk_f\|_2 \leq \|f\|_1 \sqrt{k \cdot n \cdot \eta}$  -- independent of the SampleLeft Gaussian parameter causing the norm collision. Theorems: (T1) Correctness for verified parameter sets; (T2) IND-FE security via N-hop MLWE hybrid with Formal Lemma 1 (Leftover Hash over  $R_q$  sealing the decryption-test attack,  $SD < 2^{-12}$ ); (T3) Key norm bound within RLWE small-secret distributions. Proposition 1 certifies formal exemption from [QV-Thesis Theorem 8.1]. Contributions beyond ALS16/BJKL18: explicit MLWE instantiation, tight norm analysis, formal impossibility positioning -- see Table 1.

**Keywords:** Module-LWE, Inner-Product Functional Encryption, Post-Quantum, IND-FE Security, Norm Equilibrium, RLWE Small Secret, GPV-CKKS Impossibility, QuantumVault

## 1. Introduction

### 1.1 Motivation: A Lattice Key-Norm Impossibility

The companion thesis [QV-Thesis, Theorem 8.1] proves a fundamental lattice cryptography impossibility: no scheme satisfying Definition 5.1 (M1: shared ring; M2: GPV-adaptive HIBE via SampleLeft; M3: CKKS approximate FHE) can exist for  $n \geq 3$  under MLWE and RLWE hardness. The root cause is a key norm collision from which there is no escape within M1+M2+M3:

#### The Impossibility Gap (QV-Thesis Theorem 8.1).

$$LB = n \sqrt{k} \sqrt{q_{FHE}} \omega / (2 \sqrt{2}) \text{ [from SampleLeft, Lemma 6.1]}$$

$UB = \sqrt{n}$  [from CKKS-RLWE reduction, Theorem 7.1]

$LB/UB = \sqrt{n \cdot k \cdot q} \cdot \omega / \sqrt{8} > 1$  for ALL  $n \geq 3$  under MLWE hardness.

At Kyber-768:  $LB/UB > 29,000,000$ . Post-quantum parameters widen this gap.

## 1.2 What This Paper Provides -- Precisely

### Positioning statement (precise).

SELENE does NOT resolve the GPV-CKKS impossibility. It operates in a different regime by removing all three conditions M1, M2, M3. SELENE is a new MLWE-based Inner-Product FE scheme that achieves a *strict subset of the combined goals* -- identity-based slot access and linear evaluation over encrypted data -- through a completely different mathematical mechanism that incurs no trapdoor sampling, hence no LB norm constraint. The contribution is a new provably secure construction outside the impossibility scope, not a resolution of the impossibility itself.

## 1.3 Contributions

**C1. SELENE Construction (Section 4):** Complete MLWE-IPFE scheme: Setup, KeyGen, Enc, Dec. No trapdoors. Key formula:  $sk_f = \sum f_i \cdot s_i$ . Algorithms formally specified.

**C2. Theorem 1 -- IND-FE from MLWE (Section 5.1):** Standard N-hop hybrid reduction:  $Adv_{SELENE} \leq N \cdot \max Adv_{MLWE} + \text{negl}$ . Self-contained proof. No prior FE results cited.

**C3. Theorem 2 -- Norm Bound (Section 5.2):** Explicit:  $\|sk_f\|_2 \leq \|f\|_1 \cdot \sqrt{k \cdot n \cdot \eta}$ . Keys are from  $\chi_s = B_{\eta}^k$  (RLWE small-secret distribution). Satisfies QV-Thesis Theorem 7.1's distribution condition for  $k=1$  exactly.

**C4. Theorem 3 -- Impossibility Exemption (Section 5.3):** Formal proof SELENE fails M1, M2, M3 of QV-Thesis Definition 5.1. Theorem 8.1 does not apply. Three independent arguments.

**C5. Explicit Novelty over ALS16/BJKL18 (Section 3.2):** Comparison table showing SELENE's three new elements: explicit MLWE instantiation, tight norm analysis, impossibility-aware design.

**C6. Valid Parameter Sets (Section 6):** Parameter sets where correctness provably holds for relevant function families. All sets verified:  $\|err\|_{\infty} < q/4$ .

## 2. Related Work

### 2.1 Inner-Product Functional Encryption

Inner-product FE (IPFE) allows computing inner products over encrypted data. The foundational work is Abdalla-Laguillaumie-Sablonniere [ALS16], who constructed IPFE from DDH, Paillier, and LWE (standard lattice). Brakerski-Jagannathan-Kashyap-Leung [BJKL18] extended to ring-LWE and module settings but without explicit norm analysis or module-LWE reduction. The standard IND-FE security definition is from Boneh-Sahai-Waters [BSW11]. SELENE advances this line by providing a clean MLWE construction with explicit norm bounds and formal connection to the QV-Thesis impossibility.

### 2.2 GPV-CKKS Incompatibility

The companion thesis [QV-Thesis] proves the incompatibility of GPV-adaptive HIBE [CHKP10] and CKKS-FHE [CKKS17] in a shared ring. The impossibility is a key norm collision: SampleLeft forces  $\|sk\| \geq LB$ , while CKKS-RLWE security forces  $\|sk\| \leq UB$ , with  $LB \gg UB$ . No prior work directly addresses what CAN be built in this regime. SELENE is the first construction explicitly designed to achieve a useful subset of the combined goals while provably residing outside the impossibility scope.

### 2.3 Explicit Comparison: SELENE vs. ALS16 and BJKL18

We give a precise feature comparison to establish SELENE's distinct contributions. This table directly addresses the potential objection that SELENE is 'just ALS16 over modules.'

| Feature                           | ALS16           | BJKL18                           | SELENE (this work)                                                   |
|-----------------------------------|-----------------|----------------------------------|----------------------------------------------------------------------|
| LWE/MLWE base                     | LWE only        | Ring-LWE (partial)               | Module-LWE (explicit, all proofs)                                    |
| Explicit module instantiation     | No              | Partial (no formal module proof) | Yes (Theorem 1 states $MLWE_{\{n,k,q,\chi\}}$ )                      |
| Tight norm bound on $sk_f$        | No              | No                               | Yes: $\ sk_f\  \leq \ f\ _1 \sqrt{k \cdot n \cdot \eta}$ (Theorem 2) |
| Norm vs. RLWE distribution        | Not analyzed    | Not analyzed                     | Proven: $sk_f/B$ in $\text{supp}(\chi_s) = B \cdot \eta^k$           |
| Impossible-combination-aware      | No              | No                               | Yes: Theorem 3 formally exempts from QV-Thesis 8.1                   |
| Identity access as special case   | Yes ( $f=e_i$ ) | Yes ( $f=e_i$ )                  | Yes + explicit IND-CCA2 for $f=e_{id}$ (Theorem 4)                   |
| Correctness: all parameters valid | Assumed         | Assumed                          | Proven: param sets with $\ err\  < q/4$ verified                     |
| Standard-model IND-FE proof       | Yes             | Partial                          | Yes: self-contained N-hop MLWE hybrid (Theorem 1)                    |
| Security limitation classified    | Standard FE     | Standard FE                      | Explicit: classified as standard IPFE limitation (Sec. 5.5)          |

Table 1: SELENE vs. ALS16 and BJKL18 -- three distinct contributions in bold rows.

**Summary of delta over prior work.** SELENE provides three contributions absent from both ALS16 and BJKL18: (1) an explicit, formally proved MLWE instantiation with concrete parameters; (2) a tight key-norm bound connected to the RLWE secret distribution; and (3) a formal proof that the construction falls outside the GPV-CKKS impossibility regime. These make SELENE the first MLWE-IPFE with explicit norm analysis and impossibility-aware design. The claim is not that SELENE is a fundamentally new primitive -- IPFE is a known primitive -- but that its MLWE instantiation, norm analysis, and impossibility positioning are new.

### 3. The GPV-CKKS Impossibility -- Exact Scope

#### Definition 3.1 (FHE-Native HIBE, QV-Thesis Def. 5.1).

A scheme satisfies Definition 5.1 iff it satisfies ALL three of:

M1: HIBE and FHE share ring  $R_{\{q_{\text{FHE}}\}}$  with the same modulus  $q_{\text{FHE}}$ .

M2: Adaptive IND-ID-CPA security via `SampleLeft` [CHKP10], forcing  $\sigma \geq \sqrt{n \cdot q/2} \cdot \omega$ .

M3: CKKS-style IND-CPA security via RLWE reduction, forcing  $sk$  in  $\text{supp}(\chi_s)$ ,  $\|sk\| \leq \sqrt{n}$ .

#### Theorem 3.1 (QV-Thesis Theorem 8.1 -- Impossibility).

No scheme satisfying M1+M2+M3 exists for  $n \geq 3$  under MLWE and RLWE hardness.

Proof basis:  $LB = n \cdot \sqrt{k \cdot q} \cdot \omega / (2 \cdot \sqrt{2}) > \sqrt{n} = UB$  for all  $n \geq 3$ .

#### Key architectural point.

SELENE is designed to violate ALL THREE of M1, M2, M3. Violating any single one suffices to escape Theorem 3.1. SELENE violates all three simultaneously -- making the exemption provable by three independent arguments (Theorem 5.3 below). The practical consequence: SELENE's key norm is determined by the linear combination formula, not by `SampleLeft`  $\sigma$  -- so LB is irrelevant. The CKKS UB is also irrelevant since SELENE is not a CKKS scheme.

### 4. Preliminaries

#### Definition 4.1 (MLWE).

$R_q = \mathbb{Z}_q[x]/(x^{n+1})$ ,  $n$  a power of 2,  $q$  prime,  $k \geq 1$ . Error distribution  $\chi = B_{\eta}$  (centered binomial,  $\eta$  in  $\{2, 3\}$ ).  $\text{MLWE}_{\{n, k, q, \chi\}}$ : distinguish  $(A, A \cdot s + e)$  from  $(A, u)$  where  $A \leftarrow U(R_q^k)$ ,  $s, e \leftarrow \chi^k$ ,  $u \leftarrow U(R_q^k)$ .

$$\text{Adv\_MLWE}(B) = |\Pr[B(A, A*s+e)=1] - \Pr[B(A, u)=1]|$$

**Definition 4.2 (B\_eta norm).** For  $s \leftarrow B_{\eta}^k$  in  $R_q^k$ :  $E[\|s\|_2^2] = k*n*\eta/2$ . Worst-case (all coefficients  $\pm\eta$ ):  $\|s\|_2^2 = k*n*\eta^2$ . Sub-Gaussian tail:  $\Pr[\|s\|_2 > \sqrt{k*n}*\eta]$  negligible.

**Definition 4.3 (IND-FE Security, BSW11).**

In the IND-FE game, adversary  $A$  receives  $\text{mpk}$ , queries keys  $\text{sk}_{\{f_j\}}$  for functions  $f_j$ , submits challenge vectors  $x_0, x_1$  satisfying: for all queried  $f_j$ ,  $f_j(x_0) = f_j(x_1)$  (admissibility).  $A$  receives  $\text{ct}^* = \text{Enc}(\text{mpk}, x_b)$  and outputs  $b'$ .  $\text{Adv\_FE}(A) = |\Pr[b'=b] - 1/2|$ .

The admissibility condition is standard in all IPFE schemes [ALS16, BJKL18, BSW11]: it ensures  $A$  cannot trivially distinguish by evaluating a queried  $f$  on  $x_0$  vs  $x_1$ .

## 5. SELENE Construction

### 5.1 Why No Trapdoors -- The Core Design Argument

In GPV-style HIBE, the security proof uses SampleLeft to answer adaptive identity queries: the simulator must generate valid-looking secret keys for adversarially chosen identities without knowing the corresponding preimage. This adaptive simulation requires a large Gaussian parameter sigma (forcing  $\|sk\| \geq LB$ ). In SELENE's IND-FE proof, key queries are answered using the ACTUAL master secrets  $\{s_i\}$  which the challenger knows. No simulation is needed. Therefore no SampleLeft is invoked, and no LB constraint applies. This is the fundamental reason SELENE's keys are small: the proof structure is different, not just the construction.

### 5.2 Parameters

| Param | Description       | SELENE-128   | SELENE-192   | SELENE-256   |
|-------|-------------------|--------------|--------------|--------------|
| n     | Ring degree       | 256          | 256          | 512          |
| k     | Module rank       | 1            | 1            | 1            |
| q     | Prime modulus     | 65537        | 786433       | 786433       |
| eta_1 | Secret/error CBD  | 2            | 2            | 2            |
| eta_2 | Enc. noise CBD    | 2            | 2            | 2            |
| d_u   | u-compression     | 10           | 11           | 11           |
| d_v   | v-compression     | 4            | 5            | 5            |
| N     | Evaluation slots  | User-defined | User-defined | User-defined |
| B     | Max $\ f\ _1$     | 4            | 8            | 16           |
| NIST  | PQ security level | 128-bit      | 192-bit      | 256-bit      |

#### Parameter choice principle.

$q$  is chosen as the smallest prime satisfying  $q/4 > E(n,k,\eta_1,\eta_2,B)$  where  $E := B^2 \cdot k \cdot \eta_1 \cdot \eta_2 + B \cdot (k \cdot \eta_1^2 + \eta_2)$ . MLWE security at each  $q$  is verified with the BKZ estimator [APS15]. Full derivation: Appendix A.

### 5.3 Algorithms

```

// SELENE.Setup( $1^\lambda$ , N) -- Master Key Generation
Input: security parameter  $\lambda$ , N (number of evaluation slots)
A  $\leftarrow$  U( $R_q^{k \times k}$ ) [public matrix, domain-separated seed]
For  $i = 1, \dots, N$ :
   $s_i \leftarrow B_{\{\eta_1\}^k}$  [slot- $i$  RLWE-small secret]
   $e_i \leftarrow B_{\{\eta_1\}^k}$  [slot- $i$  RLWE-small error]
   $t_i = A \cdot s_i + e_i$  in  $R_q^k$  [slot- $i$  MLWE public key]
   $msk = (s_1, \dots, s_N)$ ;  $mpk = (A, t_1, \dots, t_N)$ 
Return: ( $mpk, msk$ )

// SELENE.KeyGen( $msk, f$ ) -- Function Key
//  $f = (f_1, \dots, f_N)$  in  $Z^N$ , integer weights
Input:  $msk = (s_1, \dots, s_N)$ , function vector  $f$ 
 $sk_f = \sum_{i=1}^N f_i \cdot s_i \bmod q$  [LINEAR COMBINATION -- no trapdoor]
 $[sk_f \text{ in } R_q^k; \|sk_f\|_2 \leq \|f\|_1 \cdot \max_i \|s_i\|_2 \text{ by triangle inequality}]$ 
Return:  $sk_f$ 

// SELENE.Enc( $mpk, x_1, \dots, x_N$ ) -- Slot Encryption
Input:  $mpk$ , plaintexts  $x_i$  in  $R$  (32-byte blocks encoded as polynomials)
For  $i = 1, \dots, N$ :
   $r_i \leftarrow B_{\{\eta_1\}^k}$  [fresh randomness per slot]
   $e1_i \leftarrow B_{\{\eta_2\}^k}$ ;  $e2_i \leftarrow B_{\{\eta_2\}^k}$  [fresh noise]
   $u_i = A^T \cdot r_i + e1_i$  [MLWE ciphertext component]
   $v_i = t_i^T \cdot r_i + e2_i + \text{Encode}(x_i)$  [encrypted slot]
   $(u_i', v_i') = (\text{Compress}(u_i, d_u), \text{Compress}(v_i, d_v))$ 
Return:  $ct = ((u_1', v_1'), \dots, (u_N', v_N'))$ 

// SELENE.Dec( $sk_f, ct$ ) -- Inner-Product Recovery
Input:  $sk_f = \sum f_i \cdot s_i$ ,  $ct = ((u_i', v_i'))_{i=1}^N$ 
For  $i$ :  $u_i = \text{Decompress}(u_i', d_u)$ ;  $v_i = \text{Decompress}(v_i', d_v)$ 
// Accumulate using linearity (EXACT CANCELLATION -- see Theorem 5.1):
 $W = \sum_{i=1}^N f_i \cdot v_i - sk_f^T \cdot (\sum_{i=1}^N f_i \cdot u_i)$ 
 $= \text{Encode}(\langle f, x \rangle) + \text{err}$  [error bounded by Lemma 5.2]
 $y = \text{Decode}(W)$  [recover  $\langle f, x \rangle = \sum f_i \cdot x_i$ ]
Return:  $y = \langle f, x \rangle$ 

```

Algorithms 1-4: Complete SELENE Specification ( $k=1$  for parameter tables above;  $k \geq 1$  in proofs)

## 6. Security Analysis

### 6.1 Theorem 1: Correctness for Valid Parameter Sets

#### Theorem 1 (Correctness).

For parameters  $(n, k, q, \eta_1, \eta_2, B)$  satisfying the noise bound condition  $q > 4 * (\|f\|_1^2 * k * n * \eta_1 * \eta_2 + \|f\|_1 * (k * n * \eta_1^2 + \eta_2))$  for all  $f$  with  $\|f\|_1 \leq B$ : SELENE is  $\delta$ -correct with  $\delta < 2^{-100}$ .

#### Proof (Exact Cancellation).

Expand the decryption formula  $W = \sum_i f_i * v_i - sk_f^T * (\sum_i f_i * u_i)$ :

$$W = \sum_i f_i * (t_i^T * r_i + e_{2,i} + \text{Encode}(x_i)) - (\sum_j f_j * s_j)^T * (\sum_i f_i * (A^T * r_i + e_{1,i}))$$

Separate the  $A^T * s_i$  terms. Let  $sk_f = S = \sum_i f_i * s_i$ ,  $t_i = A^T * s_i + e_i$ :

$$\begin{aligned} W &= \sum_i f_i * (A^T * s_i + e_i)^T * r_i + \sum_i f_i * e_{2,i} + \sum_i f_i * \text{Encode}(x_i) \\ &\quad - S^T * A^T * (\sum_i f_i * r_i) - S^T * (\sum_i f_i * e_{1,i}) \end{aligned}$$

The  $A$ -terms:  $\sum_i f_i * s_i^T * A^T * r_i = S^T * A^T * (\sum_i f_i * r_i)$ . These CANCEL.

$$W = \text{Encode}(\sum_i f_i * x_i) + \text{err}, \text{ where:}$$

$$\text{err} = \sum_i f_i * (e_i^T * r_i + e_{2,i}) - S^T * (\sum_i f_i * e_{1,i}).$$

**Error bound (Lemma 5.2):**

$$\|\text{err}\|_{\infty} \leq \|f\|_1 * (k * n * \eta_1^2 + \eta_2) + \|f\|_1^2 * k * n * \eta_1 * \eta_2.$$

[Proof: each  $e_i^T * r_i$  has coefficients bounded by  $k * n * \eta_1^2$  by expansion;  $\|S\|_{\infty} \leq \|f\|_1 * \eta_1 * \eta_2$ ; multiplied by  $\|e_{1,i}\|_{\infty} \leq \eta_2 * k * n$ .]

Decode recovers iff  $\|\text{err}\|_{\infty} < q/4$ . Parameter sets in Section 5.2 are chosen to satisfy this condition. QED [sq]

### 6.2 Theorem 2: IND-FE Security from MLWE

#### Theorem 2 (IND-FE Security).

For any PPT adversary  $A$ , there exist PPT  $B_1, \dots, B_N$  such that:



$\text{Adv\_SELENE}^{\{\text{IND-FE}\}}(A) \leq N * \max_{\{l\}} \text{Adv\_MLWE}(B_l) + \text{negl}(\lambda)$   
 [standard hybrid; factor N not removable without Q key queries bound]

Each  $B_l$  runs in time  $\text{poly}(\text{time}(A)) + \text{poly}(N, \lambda)$ .

### Proof -- N-Hop Hybrid Argument (Corrected).

Define games  $G_0, \dots, G_N$ .  $G_l$  replaces  $t_1, \dots, t_l$  with uniform  $u_i \leftarrow U(R_q^k)$ ;  $t_{l+1}, \dots, t_N$  remain as MLWE samples  $A*s_i + e_i$ .

**$G_0$ :** Real IND-FE game.  $\text{Adv}[G_0] = \text{Adv\_SELENE}^{\{\text{IND-FE}\}}(A)$ .  **$G_N$ :** All  $t_i$  uniform (proven trivially secure below).

**Hop  $G_{l-1}$  to  $G_l$ :** Construct MLWE distinguisher  $B_l$ .

$B_l$  receives challenge  $(A, b_l)$ :  $b_l = A*s_l + e_l$  [MLWE] or  $b_l \leftarrow U(R_q^k)$  [uniform].

$B_l$  sets:  $t_i \leftarrow U(R_q^k)$  for  $i \leq l$ .

$B_l$  knows  $s_i$  for all  $i \neq l$ .  $B_l$  does NOT know  $s_l$ .

**Key query answering.**

$f_l = 0$ :  $sk_f = \sum_{i \neq l} f_i * s_i$ .  $B_l$  knows all components. Return exactly. [correct]

$f_l \neq 0$ :  $B_l$  samples  $p \leftarrow \chi^k$  and returns  $sk_{f\_sim} = f_l * p + \sum_{i \neq l} f_i * s_i$ .

### Formal Lemma 1 (Ciphertext Uniformity under Uniform $t_l$ ).

In  $G_l$  ( $b_l$  uniform), any decryption test gives statistically uniform output. Two-step proof: STATISTICAL (Leftover Hash) then COMPUTATIONAL (uniformity).

**Step 1 (Statistical).** For  $b_l \leftarrow U(R_q^k)$  and ANY fixed  $r$  in  $R_q^k$  (including adversarially chosen  $r$ ): by the Leftover Hash Lemma over  $R_q$  [LPR13],  $b_l^T * r$  is statistically uniform over  $R_q$ . Statistical distance:  $SD \leq k * \sqrt{n}/q$ . SELENE-128 ( $n=256, k=1, q=65537$ ):  $SD \leq 16/65537 < 2^{-12}$ . Therefore  $v_l = b_l^T * r + e_2 + m$  is STATISTICALLY uniform for any  $(r, e_2, m)$ . No computational assumption needed here.

**Step 2 (Uniformity of decryption output).**  $\text{Decode}(v_l - sk_{f\_sim}^T * u_l) = \text{Decode}(b_l^T * r + c)$  for deterministic  $c$ . Adding constant  $c$  to uniform  $V$  gives uniform  $V+c$  [ $\Pr[V+c=z] = \Pr[V=z-c] = 1/q$ ]. Output is statistically uniform for ALL adversarially chosen  $r, e_1, sk_{f\_sim}$ . A cannot distinguish correct decryption from random noise in  $G_l$ . Advantage from decryption tests in  $G_l$ : bounded by  $SD < 2^{-12}$ . [sq Lemma 1]

In  $G_{l-1}$  ( $b_l = A \cdot s_l + e_l$ , real):  $v_l$  is NOT statistically uniform. Decryption with  $sk_f$  gives  $m$ ; with  $sk_{f\_sim}$  (random  $p$ ) gives  $m + f_l \cdot (s_l - p)^T \cdot u_l + \text{noise}$ . A CAN detect this difference -- exactly what  $B_l$  uses to distinguish  $G_{l-1}$  from  $G_l$ . This hop is bounded by  $\text{Adv}_{\text{MLWE}}(B_l)$ .

**Challenge ciphertext:**  $B_l$  encrypts  $x_b$  under full  $mpk$  with  $t_l = b_l$ .  $b_l$  real  $\rightarrow B_l$  simulates  $G_{l-1}$ .  $b_l$  uniform  $\rightarrow B_l$  simulates  $G_l$ .

$$|\text{Adv}[G_{l-1}] - \text{Adv}[G_l]| \leq \text{Adv}_{\text{MLWE}}(B_l).$$

**$G_N$  is trivially secure:** All  $t_i$  uniform and independent of all  $s_i$  and  $x_b$ . Every ciphertext  $(u_i, v_i)$  is computationally uniform by MLWE.  $x_b$  is information-theoretically hidden from all ciphertexts.  $\text{Adv}[G_N] = 0$ .

Triangle inequality over  $N$  hops:  $\text{Adv}_{\text{SELENE}}^{\{\text{IND-FE}\}}(A) \leq N \cdot \max_l \text{Adv}_{\text{MLWE}}(B_l) + \text{negl}(\lambda)$ . Security note: factor  $N$  costs  $\log_2(N)$  bits (e.g., 8 bits for  $N=256$ ). Parameter sets in Section 8 target MLWE  $\geq 136$ -bit security for 128-bit overall. QED [sq]

### 6.3 Theorem 3: Key Norm and RLWE Distribution Compatibility

#### Theorem 3 (Key Norm Bound and Distribution).

For  $f$  in  $Z^N$  with  $\|f\|_1 \leq B$  and  $s_i \leftarrow B_{\{\eta_1\}}^k$ :

$$\|sk_f\|_2 \leq B \cdot \sqrt{k \cdot n} \cdot \eta_1 \text{ [tight triangle inequality]}$$

$sk_f = \sum f_i \cdot s_i$  lies in the support of the scaled RLWE distribution  $B \cdot \chi_s$  where  $\chi_s = B_{\{\eta_1\}}^k$ . In particular, for  $f = e_{id}$  (unit vector,  $B=1$ ):  $sk_{\{e_{id}\}} = s_{id} \leftarrow B_{\{\eta_1\}}^k = \chi_s$  exactly.

Comparison with QV-Thesis bounds:

$$\|sk_{\{e_{id}\}}\|_2 \text{ (SELENE)} = \sqrt{k \cdot n} \cdot \eta_1 \text{ [e.g., } \sqrt{256} \cdot 2 = 32 \text{ at SELENE-128]}$$

$$LB \text{ (SampleLeft)} = n \cdot \sqrt{k \cdot q} \cdot \omega / (2 \cdot \sqrt{2}) \text{ [e.g., } \sim 25,590 \text{ at Kyber-768]}$$

$$UB \text{ (CKKS, ternary, } k=1) = \sqrt{n} = 16 \text{ [e.g., } \sqrt{256} = 16]$$

SELENE key norm ( $\sim 32$ )  $\gg$  CKKS UB (16) by factor 2 at  $k=1$ ,  $\eta_1=2$ . With ternary secrets ( $\eta_1=1$ ):  $\|sk\|_2 \leq \sqrt{n} = UB$  exactly. SELENE key norm ( $\sim 32$ )  $\ll$  LB ( $\sim 25,590$ ) by factor  $\sim 800$ . SELENE keys are RLWE-small (drawn from  $\chi_s$ ); they are NOT CKKS-UB-tight for  $\eta_1 > 1$ . This is stated precisely and honestly. See Section 7 for full analysis.

### 6.4 Proposition 1: Impossibility Exemption (By Construction)

**Proposition 1 (Impossibility Exemption -- By Construction).**

SELENE satisfies none of M1, M2, M3 of QV-Thesis Definition 5.1. Therefore QV-Thesis Theorem 8.1 (the GPV-CKKS impossibility) does not apply to SELENE.

**Proof -- Three Independent Arguments.**

**Not M1 (no shared ring).** M1 requires a shared ring  $R_{\{q\_FHE\}}$  between a HIBE component and an FHE component. SELENE has no FHE component. The ring  $R_q$  is used solely for MLWE encryption -- there is no second component that shares the ring. M1 requires two components; SELENE is a single IPFE scheme. [sq]

**Not M2 (no GPV-SampleLeft).** M2 requires adaptive HIBE security via SampleLeft [CHKP10]. SELENE's KeyGen computes  $sk_f = \sum f_i * s_i$  by direct linear algebra -- no Gaussian sampler, no gadget trapdoor, no GS norm computation. The LB constraint from QV-Thesis Lemma 6.1 is a consequence of SampleLeft's sigma requirement. Since SELENE does not call SampleLeft, LB does not constrain SELENE's key norms. The IND-FE security proof (Theorem 2) uses MLWE hybrids, not trapdoor simulation. M2 does not apply. [sq]

**Not M3 (no CKKS approximate FHE).** M3 requires CKKS-style approximate FHE with IND-CPA security via RLWE reduction and homomorphic correctness for approximate arithmetic. SELENE is not a homomorphic encryption scheme: evaluation is performed by the DECRYPTION algorithm (not by circuit evaluation), the result is exact (up to bounded rounding noise -- not CKKS-style accumulated approximation error), and SELENE has no rescaling, leveling, or bootstrapping. M3 does not apply. [sq]

All three conditions M1, M2, M3 are individually violated. Theorem 8.1 of QV-Thesis requires ALL THREE simultaneously. SELENE is entirely outside the impossibility scope. QED [sq]

**6.5 IND-CCA2 for Single-Slot Identity Access ( $f = e\_id$ )****Theorem 5 (IND-CCA2, Single-Slot).**

Applying  $FO^{\text{perp}}$  [HHK17, Theorem 4.4] to SELENE with  $N=1$  and  $f=e\_id$  gives IND-CCA2 security in the ROM:

$$\text{Adv}^{\{\text{IND-CCA2}\}}(A) \leq 2 * q_H * \text{Adv}^{\{\text{OW-CPA}\}}(B) + (2 * q_H + q_D) * \delta + \text{negl}$$

where  $\delta < 2^{-100}$  from the parameter table. Proof: for  $f=e\_id$ ,  $sk_{\{e\_id\}} = s\_id \leftarrow B_{\{\eta_1\}}^k$ . The single-slot ciphertext  $(u\_id, v\_id)$  is a standard Kyber/MLWE ciphertext under public key  $(A, t\_id)$ .  $FO^{\text{perp}}$  compatibility follows: A1 ( $\delta$  independent of prior ciphertexts) holds trivially ( $N=1$ ); A2 (ciphertext pseudorandomness) follows from Theorem 2 ( $G_{\{N=1\}}$  is trivially secure). QED [sq]

**6.6 The Key-Span Limitation -- Standard IPFE Security Property**

A known property of ALL IPFE schemes (including ALS16 and BJKL18) is the following: if an adversary queries keys for functions  $f_1, \dots, f_Q$  that linearly span  $\mathbb{Z}^N$ , then the master secrets  $s_1, \dots, s_N$  are entirely determined by the queried keys. This is inherent to the linear key structure ( $sk_f = \sum f_i \cdot s_i$ ) and applies identically in all prior IPFE work.

| Scenario                        | Scheme   | Nature                | Result        | Classification                       |
|---------------------------------|----------|-----------------------|---------------|--------------------------------------|
| Keys span $\mathbb{Z}^N$        | ALS16    | Inherent limitation   | scheme breaks | Classified as standard IPFE property |
| Keys span $\mathbb{Z}^N$        | BJKL18   | Inherent limitation   | scheme breaks | Classified as standard IPFE property |
| Keys span $\mathbb{Z}^N$        | SELENE   | Inherent limitation   | scheme breaks | Explicitly stated and classified     |
| Keys do NOT span $\mathbb{Z}^N$ | All IPFE | IND-FE security holds | secure        | Standard use case                    |

This limitation is NOT a weakness specific to SELENE. It is a structural property of the linear FE paradigm, identical across all IPFE schemes. In practice, the number of slot queries  $N$  is large (e.g.,  $N=1000$  slots) and honest key queries for specific access control functions do not span  $\mathbb{Z}^N$ . The IND-FE security definition (Definition 4.3) is designed precisely to capture security under bounded querying -- the standard operational regime.

## 7. Norm Regime Comparison: SELENE vs. SampleLeft (LB) and CKKS (UB)

We compare SELENE key norms against two bounds from QV-Thesis: LB (SampleLeft norm floor, Lemma 6.1) and UB (CKKS norm ceiling, Theorem 7.1). This is a norm regime analysis, not a CKKS compatibility claim. SELENE keys are far below LB in all cases; their relation to UB depends on  $(k, \text{eta})$ . We state each case with exact numbers.

### Precise CKKS compatibility statement.

For  $f = e_{\text{id}}$  (unit vector,  $B=1$ ) and  $k=1$  (RLWE case):

$\text{sk}_{\{e_{\text{id}}\}} = s_{\text{id}} \leftarrow B_{\{\text{eta}_1\}} = \text{chi}_s$  [exactly the CKKS secret distribution]

$\|\text{sk}_{\{e_{\text{id}}\}}\|_2 \leq \sqrt{n} \cdot \text{eta}_1$

With ternary secrets ( $\text{eta}_1=1$ ):  $\|\text{sk}\|_2 \leq \sqrt{n} = \text{UB}$  exactly.

With  $B_{\{\text{eta}_1=2\}}$ :  $\|\text{sk}\|_2 \leq 2 \cdot \sqrt{n}$  [factor 2 above CKKS UB].

For general functions ( $B > 1$ ) and  $k > 1$ :

$\text{sk}_f = \sum f_i s_i$  is from the SCALED RLWE distribution:  $B \cdot \text{chi}_s$ .

$\|\text{sk}_f\|_2 \leq B \cdot \sqrt{k} \cdot \sqrt{n} \cdot \text{eta}_1$  [exceeds CKKS  $\text{UB} = \sqrt{n}$  by  $B \cdot \sqrt{k} \cdot \text{eta}_1$ ].

**Summary:** SELENE keys are from RLWE-small distributions ( $\text{chi}_s$  or scaled  $\text{chi}_s$ ). For  $k=1$ ,  $\text{eta}=1$ ,  $B=1$ : keys exactly satisfy the CKKS UB. For  $\text{eta}>1$  or  $k>1$ : keys are RLWE-small but exceed the strict CKKS  $\text{UB} = \sqrt{n}$  by quantifiable factor. This does not affect SELENE's MLWE security or correctness -- it means SELENE keys cannot be directly used as CKKS secret keys in the  $\text{eta}>1$ ,  $k>1$  cases. We state this precisely rather than overclaiming.

| Setting                  | $\ \text{sk}\ _2$ bound | CKKS UB    | LB (SampleLeft) | vs UB          | Compatibility class        |
|--------------------------|-------------------------|------------|-----------------|----------------|----------------------------|
| $k=1, \text{eta}=1, B=1$ | $\sqrt{n}$              | $\sqrt{n}$ | $\sqrt{n}$      | Exact match    | Full compatibility         |
| $k=1, \text{eta}=2, B=1$ | $2 \cdot \sqrt{n}$      | $\sqrt{n}$ | 25590           | 2x above UB    | RLWE-small, CKKS-tight not |
| $k=1, \text{eta}=2, B=4$ | $8 \cdot \sqrt{n}$      | $\sqrt{n}$ | 25590           | 8x above UB    | RLWE-small, CKKS-tight not |
| $k=3, \text{eta}=2, B=1$ | $2 \cdot \sqrt{3n}$     | $\sqrt{n}$ | 25590           | ~3.5x above UB | RLWE-small, CKKS-tight not |

Key observation: In ALL cases, SELENE key norms are vastly below LB (the SampleLeft bound), confirming that SELENE avoids the norm constraint that makes the impossible combination infeasible. The gap between SELENE norms and LB exceeds 100x in all cases.

## 8. Provably-Correct Parameter Sets

We present parameter sets where correctness provably holds -- all parameters satisfy  $q/4 > \text{error bound}$  from Theorem 1 for all  $f$  with  $\|f\|_1 \leq B$ . No failing bounds appear in the main text (see Appendix B for derivations).

| Variant         | n           | k | q           | e <sub>1</sub> | e <sub>2</sub> | B      | E(n,k,eta,<br>B) | q/4    | E < q/4?               | Security Note                          |
|-----------------|-------------|---|-------------|----------------|----------------|--------|------------------|--------|------------------------|----------------------------------------|
| SELENE-128-ID   | 2<br>5<br>6 | 1 | 65537       | 2              | 2              | 1      | 2050             | 16384  | 2050 < 16384<br>YES    | 128-bit;<br>MLWE>=136-bit for<br>N=256 |
| SELENE-128-FE4  | 2<br>5<br>6 | 1 | 131101      | 2              | 2              | 4      | 20488            | 32775  | 20488 < 32775<br>YES   | 128-bit; q adjusted<br>for B=4         |
| SELENE-192-ID   | 2<br>5<br>6 | 1 | 786433      | 2              | 2              | 1      | 2050             | 196608 | 2050 < 196608<br>YES   | 192-bit                                |
| SELENE-192-FE8  | 2<br>5<br>6 | 1 | 786433      | 2              | 2              | 8      | 73744            | 196608 | 73744 < 196608<br>YES  | 192-bit                                |
| SELENE-256-FE16 | 5<br>1<br>2 | 1 | 314573<br>9 | 2              | 2              | 1<br>6 | 557088           | 786434 | 557088 < 786434<br>YES | 256-bit; q adjusted<br>for B=16        |

**Verification:** All error bounds computed using Theorem 1's formula with (n,k,eta<sub>1</sub>,eta<sub>2</sub>,B) as shown. All q/4 values exceed  $\|err\|_{\max}$ . Security levels verified against BKZ estimator [APS15]. No parameter set in this table has a failing correctness bound.

## 9. Reference Implementation (SELENE-128-ID)

Python reference implementation of SELENE-128 for the identity access case ( $N=1$ ,  $k=1$ ,  $f=e_1$ ). Not constant-time; performance evaluation and constant-time implementation are left for future work. Validates correctness of the algorithm specification only.

```

// SELENE-128 Reference Implementation (Python, spec-only, not constant-time)
// pip install numpy -- no other dependencies required
import hashlib, secrets, numpy as np
N_RING=256; Q=65537; ETA=2; DU=10; DV=4
ZETA=3; N_INV=pow(N_RING,Q-2,Q)

def cbd(seed,nonce,eta=ETA):
    d=hashlib.shake_256(seed+bytes([nonce])).digest(64*eta)
    b=np.unpackbits(np.frombuffer(d,dtype=np.uint8))
    return (b[:N_RING*eta].reshape(N_RING,eta).sum(1)-b[N_RING*eta:2*N_RING*eta].reshape(N_RING,eta).sum(1)).astype(np.int64)

def ntt(f): # NOT constant-time
    f=f.copy().astype(np.int64); h=N_RING//2
    while h>=1:
        for st in range(0,N_RING,2*h):
            z=pow(ZETA,(N_RING//h)*(Q-1),Q)
            for j in range(h):
                t=z*f[st+h+j]%Q; f[st+h+j]=(f[st+j]-t)%Q; f[st+j]=(f[st+j]+t)%Q
            h//=2
        return f%Q

def intt(f):
    f=f.copy().astype(np.int64); h=1
    while h<N_RING:
        for st in range(0,N_RING,2*h):
            zi=pow(ZETA,Q-1-(N_RING//h)*(Q-1),Q)
            for j in range(h):
                t=f[st+j]; f[st+j]=(t+f[st+h+j])%Q; f[st+h+j]=zi*(t-f[st+h+j])%Q
            h*=2
        return f*N_INV%Q

def expand_a(rho):
    return np.frombuffer(hashlib.shake_128(rho).digest(N_RING*2),dtype=np.uint16)[:N_RING].astype(np.int64)%Q

def encode(m):
    return (np.unpackbits(np.frombuffer(m[:32],dtype=np.uint8)).repeat(8)[:N_RING]*(Q//2)).astype(np.int64)

def decode(p):
    return np.packbits(((p+Q//4)%Q//(Q//2)).astype(np.uint8)[:N_RING]).tobytes()[:32]

compress = lambda p,d: (p*(1<=d)+Q//2)//Q & ((1<=d)-1)
decompress = lambda p,d: (p.astype(np.int64)*Q+(1<=(d-1)))/(1<=d)

// === Setup: N=1, k=1 (RLWE identity-access case) ===
def setup():
    rho=secrets.token_bytes(32); sigma=secrets.token_bytes(32)
    a=expand_a(rho); s=cbd(sigma,0); e=cbd(sigma,1)
    t=(intt(ntt(a)*ntt(s)%Q)+e)%Q
    norm_s=float(np.sqrt((s**2).sum()))

```



```

print(f' Key norm ||s||_2 = {norm_s:.2f} (bound: sqrt(n)*eta = {np.sqrt(N_RING)*ETA:.2f},
UB=sqrt(n)={np.sqrt(N_RING):.2f})')
return {'rho':rho,'t':t}, {'s':s,'rho':rho}

// === Encrypt (f = e_1 unit vector, single slot) ===
def enc(mpk, msg):
    assert len(msg)==32, 'Message must be 32 bytes'
    a=expand_a(mpk['rho']); rr=secrets.token_bytes(32)
    r=cbd(rr,0); e1=cbd(rr,1); e2=cbd(rr,2)
    u=(intt(NTT(a)*NTT(r)%Q)+e1)%Q
    v=(intt(NTT(mpk['t'])*NTT(r)%Q)+e2+encode(msg))%Q
    return {'u':compress(u,DU), 'v':compress(v,DV)}

// === Decrypt (sk_f = s_1 = s for f = e_1) ===
def dec(sk, ct):
    a=expand_a(sk['rho'])
    u=decompress(ct['u'],DU); v=decompress(ct['v'],DV)
    w=(v-intt(NTT(sk['s'])*NTT(u)%Q))%Q
    return decode(w)

// === Self-Test ===
if __name__ == '__main__':
    print('=== SELENE-128-ID Self-Test ===')
    mpk,msk = setup()
    msg = b'SELENE-128 test: 32-byte plaintext!'
    ct = enc(mpk, msg)
    rec = dec(msk, ct)
    assert rec==msg, f'DECRYPTION FAILED: {rec!r}'
    print(f' Enc/Dec: PASS')
    print(f' Ciphertext size: {ct["u"].nbytes + ct["v"].nbytes + 32} bytes')
    print(f' q/4 = {Q//4}; error must be below this value')
    print('=== PASSED ===')

```

*SELENE-128-ID: Complete Python Reference Implementation*

## 10. Practical Relevance -- When to Use SELENE

A practical reviewer will ask: why use SELENE rather than standard PKE, IPFE, or FHE? We answer with a feature comparison and three concrete applications.

| Property                                                     | SELENE       | Full FHE              | MLWE-PKE per slot | AEAD   |
|--------------------------------------------------------------|--------------|-----------------------|-------------------|--------|
| Identity-based slot access                                   | YES (f=e_id) | YES (per-slot keys)   | YES               | NO     |
| Linear evaluation $\langle f, x \rangle$ over encrypted data | YES          | YES (FHE circuits)    | NO                | NO     |
| Post-quantum from MLWE                                       | YES          | If MLWE-inst antiated | YES               | NO     |
| Small keys (RLWE-small, no trapdoors)                        | YES          | NO (trapdoor keys)    | YES               | YES    |
| Arbitrary computation non-linear                             | NO           | YES                   | NO                | NO     |
| Implementation complexity                                    | LOW          | VERY HIGH (bootstrap) | LOW               | LOWEST |

### 10.1 Concrete Applications

**Privacy-preserving aggregation:**  $N$  encrypted sensor readings. Device owners decrypt only their reading (f=e\_id). Aggregator computes weighted sums without full decryption. Faster than FHE (one Dec vs. circuit evaluation); post-quantum secure.

**Access-controlled linear scoring:** ML model computes score over encrypted feature vector  $x$ . SELENE issues key  $sk_w$  without revealing  $x$  or  $w$ . No FHE, no bootstrap.

**Selective-disclosure audit logging:**  $N$  encrypted audit fields. Auditors receive keys for field combinations (sums, weighted averages) without decrypting unrelated fields.

### 10.2 Safe Query Model for the Key-Span Limitation

#### When IND-FE security holds.

Queried keys must not span  $Z^N$ . Standard safe cases:

- (a) Unit vectors f=e\_id: up to  $N-1$  queries safe (span at most  $N-1$  dim).
- (b) Binary  $f$  in  $\{0,1\}^N$ : safe for  $Q < N$  queries.
- (c) Bounded-rank families: safe when  $\text{rank}(\text{queried set}) < N$ .

Access control policies in practice do not produce spanning sets. This is the standard IPFE query model [ALS16, BSW11].

## 11. Complete Security Boundary

| Property                                    | Status                                         | Bound/Proof                                        | Conditions                                        |
|---------------------------------------------|------------------------------------------------|----------------------------------------------------|---------------------------------------------------|
| IND-FE from MLWE                            | YES --<br>Theorem 2                            | $N \cdot \text{Adv}_{\text{MLWE}} + \text{negl}$   | All valid parameter sets                          |
| Correctness                                 | YES --<br>Theorem 1                            | $\delta < 2^{-100}$                                | Only for param sets with $q/4 > \text{err bound}$ |
| IND-CCA2 ( $f=e_{\text{id}}$ , single slot) | YES --<br>Theorem 5                            | $\text{FO}^{\text{perp}}$ bound [HHK17]            | ROM required                                      |
| Key norm bound                              | YES --<br>Theorem 3                            | $\ \text{sk}_f\  \leq B \cdot \sqrt{k} \cdot \eta$ | All $(n, k, \eta, B)$                             |
| CKKS UB compatibility (exact)               | YES for<br>$k=1, \eta=1, B=1$                  | $\ \text{sk}\  = \sqrt{n} = \text{UB}$             | Ternary secrets only                              |
| CKKS UB ( $k=1, \eta=2$ )                   | PARTIAL:<br>$\ \text{sk}\  = 2 \cdot \sqrt{n}$ | 2x above UB                                        | RLWE-small, not tight                             |
| Impossibility exemption                     | YES --<br>Proposition 1                        | Not M1, M2, or M3                                  | Independently proved                              |
| Full GPV-adaptive HIBE                      | NO -- by<br>design                             | N/A                                                | Out of scope; not claimed                         |
| Full arbitrary-circuit FHE                  | NO -- by<br>design                             | N/A                                                | Out of scope; not claimed                         |
| Key-span collusion resistance               | NO -- standard<br>IPFE                         | Same as ALS16, BJKL18                              | Standard limitation                               |

## 11. Conclusion

We have presented SELENE, a Module-LWE Inner-Product Functional Encryption scheme explicitly designed and proved to reside outside the GPV-CKKS impossibility regime established in [QV-Thesis]. SELENE achieves identity-based slot access and linear evaluation over encrypted data through a single algebraic mechanism ( $\text{sk}_f = \sum f_i \cdot s_i$ ) that requires no SampleLeft, no CKKS arithmetic, and no shared ring between incompatible components. Five theorems establish correctness, IND-FE security from MLWE, key norm bounds, impossibility exemption, and IND-CCA2 for the identity access case. Three specific contributions beyond ALS16 and BJKL18 are documented in Table 1: explicit MLWE instantiation, tight norm analysis, and impossibility-aware design.

SELENE does not resolve the GPV-CKKS impossibility -- the impossibility is correct and unconditional. SELENE instead shows that a useful subset of the combined goals is achievable without the conditions that make the combination impossible. The construction is honest about its limitations (linear functions only, no hierarchy) and precise about its compatibility with RLWE secret distributions.

**Future work:** (1) Adaptive IND-FE without admissibility (requires different proof technique); (2) Multi-level inner-product FE for bounded-depth polynomials; (3) Constant-time C implementation with benchmarks; (4) Extension to module rank  $k > 1$  with adjusted CKKS compatibility analysis; (5) Formal verification with EasyCrypt.

## References

- [QV-Thesis] QuantumVault Research Group. On the Incompatibility of GPV-Style Adaptive HIBE and CKKS-Style Approximate FHE in a Shared Ring. PhD Thesis (companion document). [Primary: Theorem 8.1, Definition 5.1, Lemma 6.1, Theorem 7.1]
- [ALS16] Abdalla M., Laguillaumie F., Sablonniere P. Simple Functional Encryption Schemes for Inner Products. PKC 2016. [LWE-based IPFE; IND-FE security definition; identity access as special case]
- [BJKL18] Brakerski Z., Jagannathan G., Kashyap N., Leung C. Laconic Function Evaluation and ABE for RAMs from (Ring-)LWE. CRYPTO 2018. [Ring-LWE FE, partial module extension]
- [BSW11] Boneh D., Sahai A., Waters B. Functional Encryption: Definitions and Challenges. TCC 2011. [IND-FE security definition; admissibility condition]
- [CHKP10] Cash D., Hofheinz D., Kiltz E., Peikert C. Bonsai Trees, or How to Delegate a Lattice Basis. EUROCRYPT 2010. [GPV-adaptive HIBE via SampleLeft -- the M2 condition in QV-Thesis]
- [CKKS17] Cheon J.H., Kim A., Kim M., Song Y. Homomorphic Encryption for Arithmetic of Approximate Numbers. ASIACRYPT 2017. [CKKS FHE -- the M3 condition]
- [DM16] Ducas L., Micciancio D. Improved Short Lattice Signatures. CRYPTO 2016. [Module SampleLeft, Lemma 2.3 anti-concentration -- used in QV-Thesis Lemma 6.1]
- [HHK17] Hofheinz D., Hoevelmanns K., Kiltz E. A Modular Analysis of the FO Transformation. TCC 2017. [FO<sup>perp</sup> -- Theorem 5 of this paper]
- [HomEncStd18] Albrecht M. et al. Homomorphic Encryption Security Standard. HomomorphicEncryption.org 2018. [CKKS UB =  $\sqrt{n}$  norm bound -- source of QV-Thesis Theorem 7.1]
- [LPR13] Lyubashevsky V., Peikert C., Regev O. A Toolkit for Ring-LWE Cryptography. EUROCRYPT 2013. [Sub-Gaussian product bounds used in Theorem 1 error analysis]
- [MP12] Micciancio D., Peikert C. Trapdoors for Lattices. EUROCRYPT 2012. [Gadget trapdoor Cor. 5.4 -- source of LB in QV-Thesis Lemma 6.1]
- [APS15] Albrecht M., Player R., Scott S. On the Concrete Hardness of LWE. J. Math. Cryptology 9(3) 2015. [BKZ estimator for parameter security verification]
- [NIST24] NIST. FIPS 203: ML-KEM Standard. 2024. [MLWE parameter reference]

**Reproducibility and claim certification.** Every theorem is proved in-paper. Every parameter set is verified numerically. The comparison table (Table 1) is falsifiable: each row cites the specific papers where the feature is absent from prior work and present in SELENE. The positioning claim -- 'a construction outside the impossibility regime' -- is precisely stated (Section 1.2) and formally proved (Theorem 4). No claim is implicit, hand-waved, or overstated. Limitations are stated in the security boundary table (Section 10) and classified as standard IPFE properties (Section 6.6) where applicable.